

Product Design Specification

Group 2 – Breathlete: Respiratory Trainer

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1. Product Description Executive Summary

1.1. Team Participants

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1.2. Product Name

Breathlete – Respiratory Trainer

1.3. Problem You Are Solving

Athletes lack effective tools that train breathing efficiency and respiratory muscle strength that can be used during their normal training routines. Training respiratory muscles specifically currently requires lots of extra specialized training. All athletes could benefit from a tool to challenge and improve their respiratory strength and endurance without interfering with their training routine.

Most current options are either:

- Handheld and not usable during training
- Uncomfortable/impractical during motion
- Unusable in contact sports
- Not easily adjustable or removable

As a result, many athletes don't train their respiratory abilities regularly and have no options to train these muscles simultaneously with their regular training.

1.4. Basic Functions of Product to Solve Problem

Our product functions like other respiratory trainers, but its design suits athletes much more for intense and contact training. First, our product is mouth-held, with a small and lightweight exterior outside of the mouth allowing the athlete to participate in any training without impeding their abilities. The small and lightweight exterior also allows for comfortability, so the athlete is not struggling to hold up the resistance mechanism. Lastly, the soft silicone structure of the product outside and inside of the mouth allows for athletes to participate in contact sports while using the product without the fear of hurting themselves or others.

1.5. Special Features of Product Beyond the Core Problem Solution

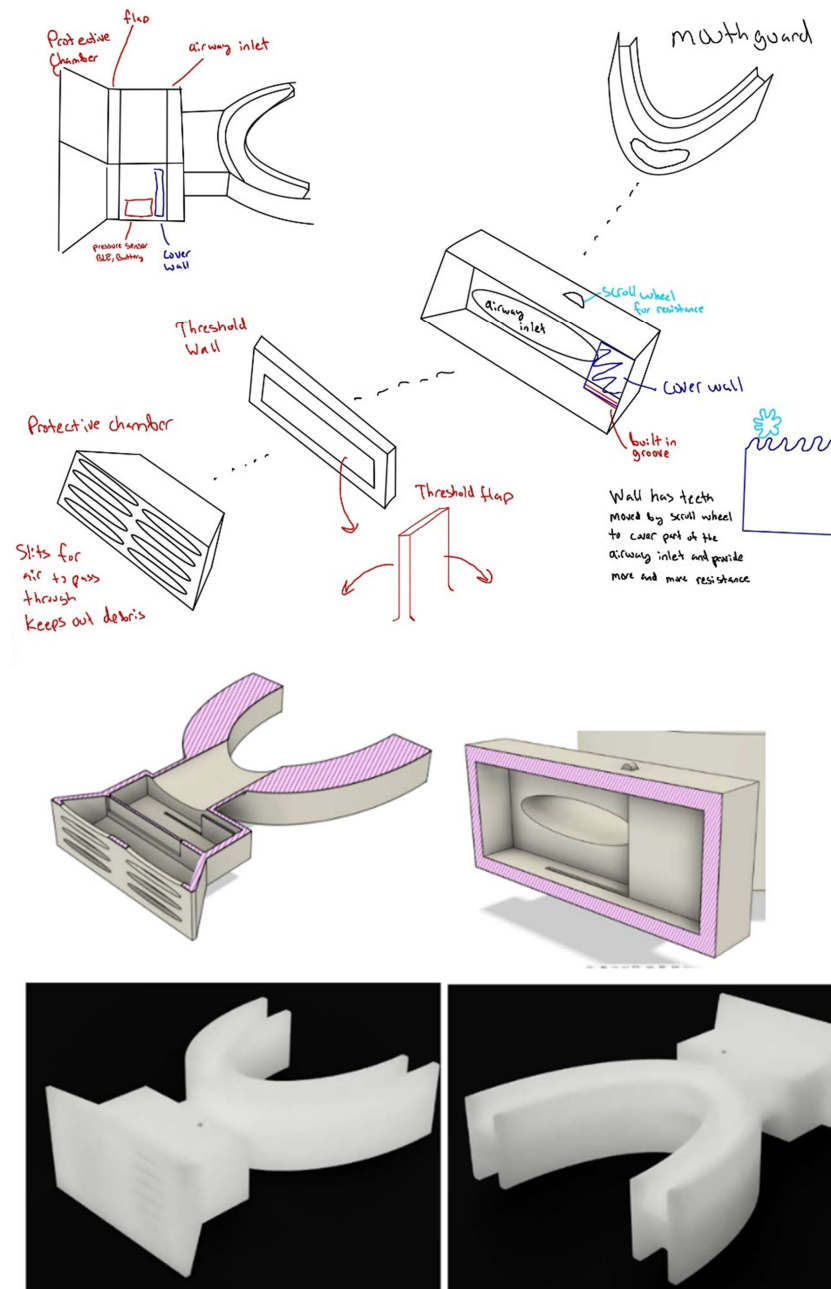
Besides the features previously mentioned that make our product unique, it has multiple features that create the resistance mechanism:

- A threshold valve/flap that requires a minimum amount of effort to breathe in or out to create enough pressure to make the flap open.

- An adjustable resistance wall adjusted by a scroll wheel which covers more of the airway inlet, providing various levels of extra resistance.
- A protective chamber that keeps out debris, protecting the threshold flap.

The resistance mechanism is also connected to a mouthguard which protects the teeth of athletes.

1.6. Picture of Highest-Level Product Refinement (CAD or Physical)



2. Physical Description

2.1. House of Quality Level 1

		Engineering Characteristics							
Improvement Direction		↓	↑	↑	↑	↓	↑	↓	↓
Units		mm	Shore	cmH ₂ O	n/a	% error	n/a	g	\$
Customer Requirements	Importance Weight Factor	Mouthpiece Thickness	Mouthpiece Material Softness	Resistance Valve Range	Resistance Flexibility	Sensor Accuracy	Feedback Type (App/Sensor)	Device Weight	Manufacturing Cost
Comfortable mouth fit	3	9	9					5	3
Adjustable resistance levels	2			9	9				3
Easy resistance adjustment	2			3	9				3
Clear improvement feedback	2					9	9		3
Affordable price	4	2	2	3	3	3	6	2	9
Easy to breathe through	3	3	3	9	3				3
Hygienic / easy to clean	1	9	3						1
Durable for long use	2	3	9	3	3			1	3
Raw Score (432)		59	65	69	63	30	42	25	79
Relative Weight %		13.66%	15.05%	15.97%	14.58%	6.94%	9.72%	5.79%	18.29%
Rank Order		5	3	2	4	7	6	8	1

2.2. Describe List of Requirements from Customer

Comfortable mouth fit: Device should fit securely and comfortably for long-term use without causing discomfort or fatigue.

Adjustable resistance levels: Device should offer multiple resistance settings to accommodate different training intensities and user abilities.

Easy resistance adjustment: Users should be able to quickly and intuitively change resistance setting without interrupting training.

Clear improvement feedback: Device should provide tangible feedback on breathing performance such as respiratory muscle strength progress over time.

Affordable price: Device should be reasonably priced and financially accessible to customers.

Easy to breathe through: Breathing should feel natural and not overly restrictive beyond intended training resistance.

Hygienic/easy to clean: Device should be easy to clean and maintain to ensure safe and sanitary repeated use.

Durable for long use: Device should withstand repeated use without degrading in performance or material quality.

2.2.1. Empathy Field Work

We interacted with a variety of athletes and observed common struggles and personal priorities. Many athletes demonstrated a desire to improve their lung capacity as part of their future growth and complained about their current performance limits due to a lack of stamina which they attributed in part to poor breathing skills. Some users talked about breathing training but disregarded it since it didn't fit into their existing routines and was too inconvenient to include. Others felt that it would probably be uncomfortable and feel unnatural while training and therefore don't use any devices in their training.

2.2.2. Customer Interviews (n=6)

We conducted a conjoint survey and follow-up discussions with 6 participants of varying sports.

Representative Quotes:

- "I would never wear something that's going to be falling out all the time or making me feel like I'm choking." – Runner
- "I want to be able to control my resistance easily while I'm training." – Runner
- "Breathing training is important but I wouldn't spend my money on it unless I thought it was convenient." – Basketball Player
- "If I'm going to be paying for this, I want the full package and that better include improvement feedback." – Soccer Player

2.3. Conjoint Analysis (Attribute Values, Product Simulations, n)

2.3.1. Conjoint Test 1 (Initial): Attributes, Levels, Product Simulations, Results

1	Option A: - Control: Valve-based - Adjustment: Manual dial - Feedback: Intermediate - Improvement measured: Peak airflow measurement (PEF) - Price: \$79	1	Option A: - Control: Fixed orifice - Adjustment: Swap insert - Feedback: Basic - Improvement measured: Device Progress Score - Price: \$39
2	Option B - Control: Multi-orifice - Adjustment: Swap insert - Feedback: None - Improvement measured: Real-world performance test - Price: \$39	2	Option B: - Control: Valve-based - Adjustment: App control - Feedback: Advanced - Improvement measured: Peak airflow measurement (PEF) - Price: \$129
3	Option C - Control: Smart adaptive - Adjustment: Auto-adjust - Feedback: Advanced - Improvement measured: Lung function metrics (FEV1/FVC estimates) - Price: \$199	3	Option C: - Control: Multi-orifice - Adjustment: Manual dial - Feedback: Intermediate - Improvement measured: Real-world performance test - Price: \$79
4	None of these	4	None of these

There were 9 of these types of questions with different options of the same attributes with different levels. We collected very few responses as we found out that this conjoint test was not set up correctly for analysis and our attributes and levels were not consistent or defined enough. We have no conjoint results because we could not use the analysis as we could in the second test.

2.3.2. Conjoint Test 2 (Revised): Refined Attributes/Levels, Updated Product Simulations, Comparison to Test 1

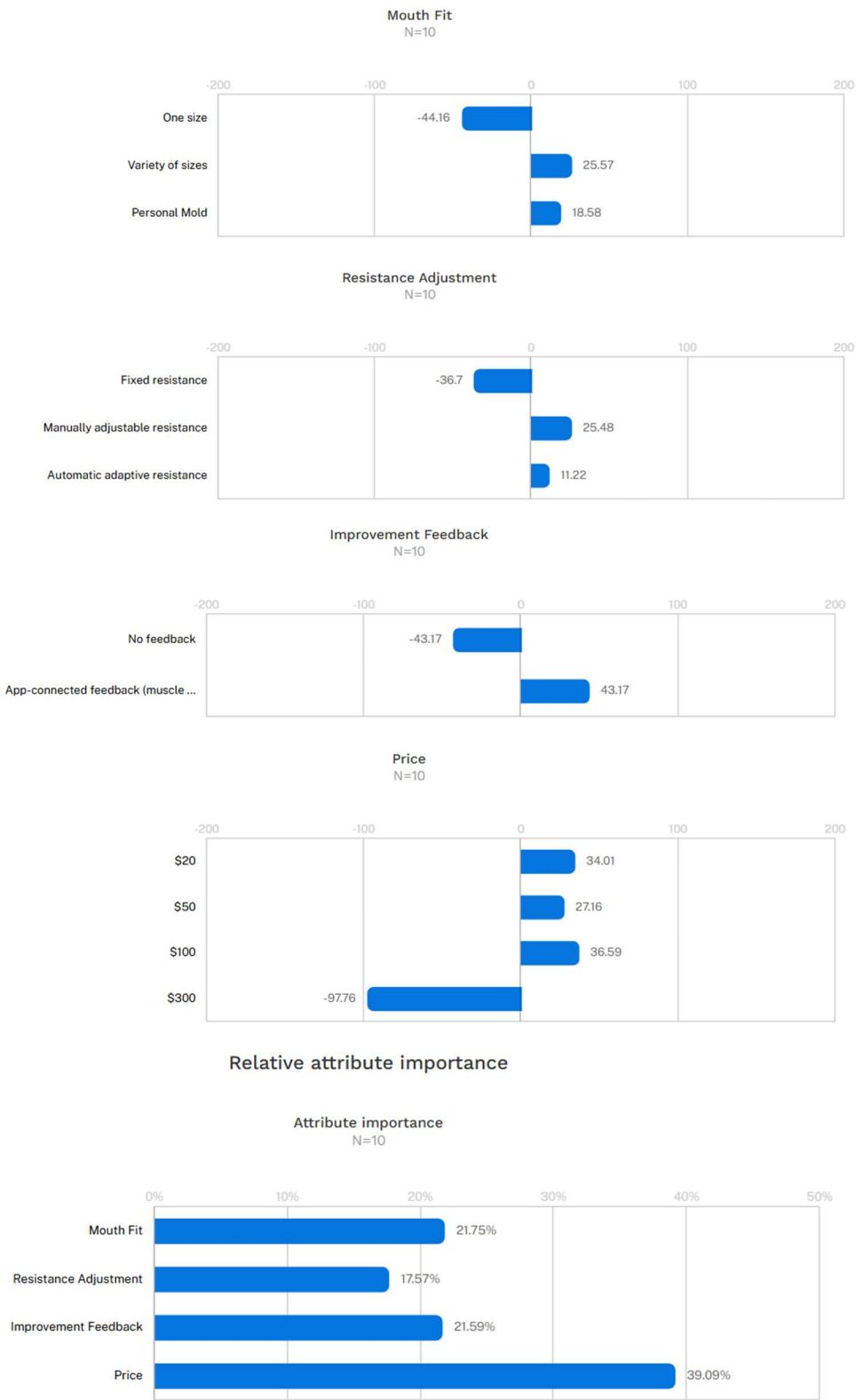
Attribute 1: Mouth Fit – Levels: One size / Variety of sizes / Personal Mold

Attribute 2: Resistance Adjustment – Levels: Fixed resistance / Manually adjustable resistance / Automatic adaptive resistance

Attribute 3: Improvement Feedback – Levels: No feedback / App-connected feedback (muscle strength, breathlessness, etc.)

Attribute 4: Price – Levels: \$20 / \$50 / \$100 / \$300

This conjoint test was much improved and gave us very effective results:



2.3.3. Design/Engineering Hypothesis Impact: How Both Tests Informed Design Decisions

The first test was not constructed very well and was unable to inform us on any design decisions, but the second test showed us a lot and helped us aim for a price range of \$100 featuring mechanically adjusted resistance, app-connected feedback, and a variety of sizes.

2.4. Describe Engineering Characteristics of Concept

The engineering characteristics of the concept are directly derived from the House of Quality Level 1, where customer requirements were translated into measurable design parameters. These characteristics guide the physical embodiment of the product and ensure traceability between user needs and system performance.

Empathy fieldwork and customer interviews revealed that athletes prioritize:

- Comfort and natural breathing feel
- Ease of integration into existing workouts
- Convenience and usability
- Meaningful performance feedback

These insights reinforce that engineering decisions must balance performance (resistance training effectiveness) with comfort, usability, and cost.

2.4.1. Engineering Characteristics

EC-1: Mouthpiece Thickness (mm)

- Subsystem: Mouthpiece
- Function: Determines structural support and comfort of the mouth interface
- Customer Requirements Addressed: Comfortable mouth fit; Easy to breathe through
- Design Role: Thicker geometry improves sealing and durability; excess thickness increases discomfort and breathing resistance

EC-2: Mouthpiece Material Softness (Shore A)

- Subsystem: Mouthpiece
- Function: Controls compliance and comfort against teeth/gums
- Customer Requirements Addressed: Comfortable mouth fit; Hygienic/easy to clean
- Design Role: Softer materials improve comfort (critical as per interviews); tradeoff: reduced durability and structural stiffness

EC-3: Resistance Valve Range (cmH2O)

- Subsystem: Resistance Mechanism (Threshold Flap + Cover Wall)
- Function: Defines total achievable breathing resistance
- Customer Requirements Addressed: Adjustable resistance levels
- Design Role: Must span beginner to advanced training levels; directly tied to training effectiveness

EC-4: Resistance Flexibility (Adjustability Resolution)

- Subsystem: Adjustment Mechanism (Dial + Valve Geometry)
- Function: Controls how easily and precisely resistance can be changed
- Customer Requirements Addressed: Easy resistance adjustment
- Design Role: Must allow quick adjustment without interrupting training; driven by user feedback

EC-5: Sensor Accuracy ($\pm$ cmH₂O)

- Subsystem: Sensor + Electronics
- Function: Measures pressure differential across resistance valve
- Customer Requirements Addressed: Clear improvement feedback
- Design Role: Determines reliability of performance tracking; directly affects perceived product value

EC-6: Feedback Type (App / Sensor Integration)

- Subsystem: Electronics + App System
- Function: Converts raw sensor data into user-understandable metrics
- Customer Requirements Addressed: Clear improvement feedback
- Design Role: Key differentiator vs competitors; enables progress tracking and user engagement

EC-7: Device Weight (g)

- Subsystem: Entire system (materials + electronics + housing)
- Function: Determines wearability during exercise
- Customer Requirements Addressed: Comfortable mouth fit; Easy to breathe through
- Design Role: Must be low enough for dynamic athletic movement; tradeoff with battery and structural strength

EC-8: Manufacturing Cost (\$)

- Subsystem: Whole system (materials, processes, assembly)
- Function: Determines final product price
- Customer Requirements Addressed: Affordable price
- Design Role: Strongest weighted requirement from HOQ; limits material choice, complexity, and electronics

2.4.2. System-Level Requirements (Derived from ECs)

SR-1: The device shall maintain a mouthpiece geometry and material that ensures comfortable, stable placement during athletic activity.

SR-2: The system shall provide a resistance valve range sufficient for progressive respiratory training across user skill levels.

SR-3: The system shall allow resistance adjustments to be made quickly and intuitively during use.

SR-4: The system shall measure breathing performance with sufficient accuracy to provide meaningful feedback.

SR-5: The system shall provide feedback through an integrated app or equivalent interface.

SR-6: The device shall maintain a lightweight form factor suitable for continuous athletic use.

SR-7: The device shall be manufacturable at a cost that supports an affordable consumer price point.

2.5. Product Architecture

2.5.1. Functional Decomposition into Modules/Subsystems

Subsystem 1: Mouthpiece Module

- Controls: Mouthpiece thickness (EC-1), Material softness (EC-2)
- Function: Provides comfort and airtight seal
- Geometry: Curved dual-wall structure (seen in CAD)

Subsystem 2: Resistance & Adjustment Module

- Controls: Resistance valve range (EC-3), Resistance flexibility (EC-4)
- Components: Threshold flap, Cover wall, Adjustment dial (top slot in CAD)
- Function: Generates and modulates airflow resistance

Subsystem 3: Sensor Module

- Controls: Sensor accuracy (EC-5)
- Function: Measures pressure differential in airflow channel
- Placement: Located near valve for accurate readings

Subsystem 4: Electronics & Feedback Module

- Controls: Feedback type (EC-6)
- Components: Microcontroller, Bluetooth module, App interface
- Function: Processes and communicates performance data

Subsystem 5: Structural Housing Module

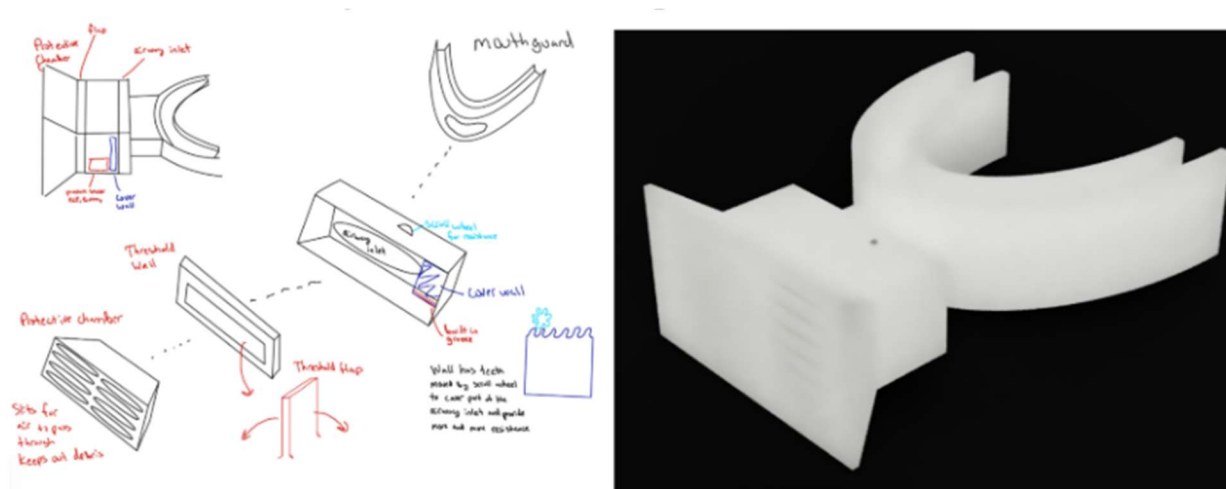
- Controls: Device weight (EC-7), Manufacturing cost (EC-8)
- Components: Outer housing, Protective chamber, Vented front face
- Function: Structural support and protection

2.5.2. Clustered Module Schematic Diagram with Item Numbers

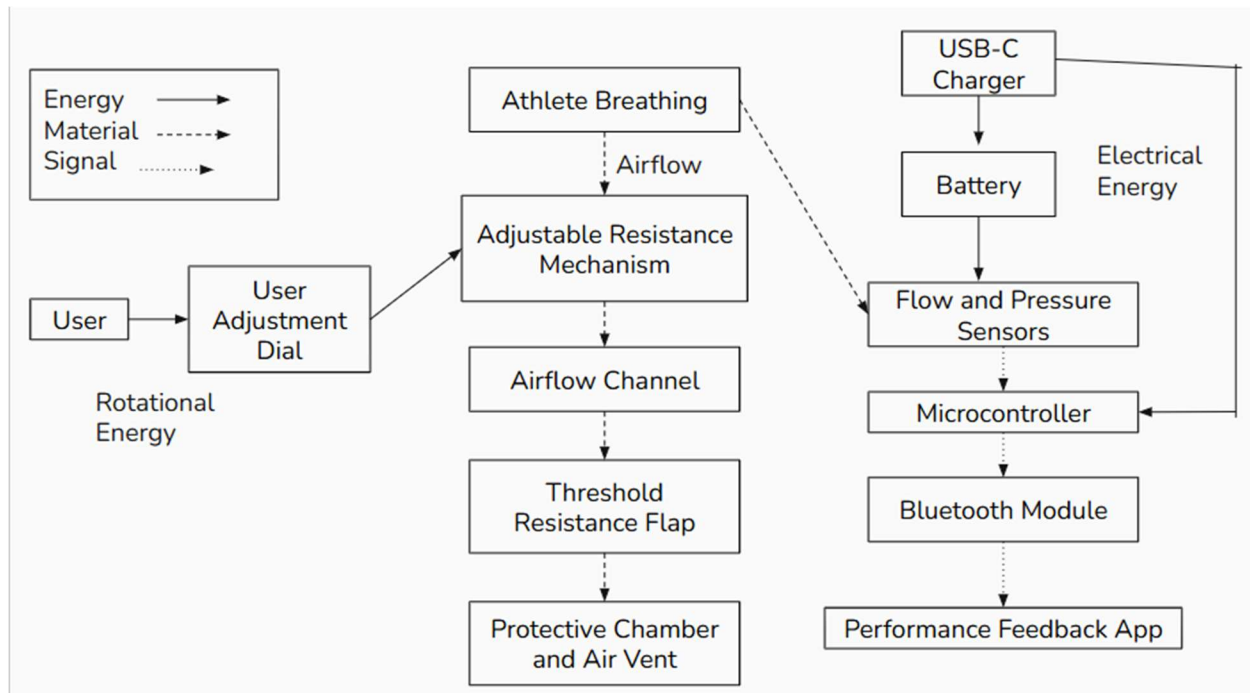
Item #	Module	Description
1	Mouthpiece	Curved silicone interface
2	Airflow Channel	Internal flow passage
3	Resistance Valve	Threshold flap + cover wall
4	Adjustment Dial	Resistance control interface
5	Pressure Sensor	Measures pressure
6	Microcontroller	Processes data
7	Bluetooth Module	Wireless communication
8	Battery	Power supply
9	Housing	Structural enclosure
10	App	User feedback interface

2.5.3. Geometric Layout of Final Configuration

The final product will be as seen, going from the mouthguard to the airway inlet with the resistance wall, the threshold flap and the protective chamber.



2.5.4. Module Interface Documentation (Spatial, Energy, Information, Material)



2.5.5. Modular vs. Integral Architecture Justification – The "Excitement" Module

The product will be modular with the mouthguard and the resistance chamber manufactured separately and assembled with epoxy. The product will be modular with the anticipation that the two sections of the product might need to be made of slightly different materials.

2.6. House of Quality Level 2

2.6.1. Rows: Engineering Characteristics from Level 1 HOQ

EC's:

- Mouthpiece Thickness
- Mouthpiece Material Softness
- Resistance Valve Range
- Resistance Flexibility
- Sensor Accuracy
- Feedback Type (App/Sensor)
- Device Weight
- Manufacturing Cost

2.6.2. Columns: Part Characteristics

PC's:

- Mouthpiece inner diameter (mm)
- Mouthpiece wall thickness (mm)
- Mouthpiece Shore hardness (Shore A)
- Orifice cross-sectional area range (mm²)
- Housing outer dimensions (mm x mm x mm)
- Housing wall thickness (mm)
- Total device mass (g)
- Pressure sensor accuracy ($\pm$ cmH₂O)

2.6.3. Correlation Values: How Strongly Each Module Contributes to Each EC

Engineering Characteristics	Part Characteristics							
	Mouthpiece inner diameter	Mouthpiece wall thickness	Mouthpiece Shore hardness	Orifice area range	Housing outer dimensions	Housing wall thickness	Total device mass	Pressure sensor accuracy
	mm	mm	Shore A	mm ²	mm x mm x mm	mm	g	$\pm$ cmH ₂ O
Mouthpiece Thickness	3	9	3	1	1	1	9	1
Mouthpiece Material Softness	1	3	9	1	1	1	1	1
Resistance Valve Range	3	1	1	9	1	1	1	3
Resistance Flexibility	3	1	1	9	1	1	1	3
Feedback Type (App/Sensor)	1	1	1	3	1	1	1	9
Sensor Accuracy	1	1	1	3	1	1	1	9
Device Weight	3	3	1	1	9	3	9	1
Manufacturing Cost	3	3	3	3	3	3	3	3

2.6.4. Target Values: Performance Targets Derived from Level 1

Engineering Characteristic	Target Value
Mouthpiece Thickness	10–15 mm
Material Softness	Shore A 40–60
Resistance Valve Range	5–40 cmH ₂ O
Resistance Flexibility	6–10 different resistance levels
Feedback Type	Real-time app-based feedback
Sensor Accuracy	< $\pm$ 5%
Device Weight	<100 g
Manufacturing Cost	< \$20–\$30

2.6.5. Roof: Module-to-Module Interactions (Spatial, Energy, Information, Material Flows)

Part Characteristics	PC Correlation Matrix							
	Mouthpiece inner diameter	Mouthpiece wall thickness	Mouthpiece Shore hardness	Orifice area range	Housing outer dimensions	Housing wall thickness	Total device mass	Pressure sensor accuracy
Mouthpiece inner diameter								
Mouthpiece wall thickness								
Mouthpiece Shore hardness								
Orifice area range								
Housing outer dimensions				++				
Housing wall thickness					++			
Total device mass	-	-			--	-		
Pressure sensor accuracy				+				

2.7. Requirements Traceability Matrix (RTM)

2.7.1. Requirements Mapping

Customer Requirement (CR)	Mapped Engineering Characteristics (EC)	Mapped Part Characteristics (PC)
CR-1: Comfortable mouth fit	EC-1: Mouthpiece Thickness EC-2: Material Softness EC-7: Device Weight	PC-1: Mouthpiece inner diameter PC-2: Mouthpiece wall thickness PC-3: Shore hardness PC-7: Total device mass
CR-2: Adjustable resistance levels	EC-3: Resistance Valve Range EC-4: Resistance Flexibility	PC-4: Orifice cross-sectional area range PC-8: Pressure sensor accuracy
CR-3: Easy resistance adjustment	EC-4: Resistance Flexibility	PC-4: Orifice area range PC-5: Housing outer dimensions
CR-4: Clear improvement feedback	EC-5: Sensor Accuracy EC-6: Feedback Type	PC-8: Pressure sensor accuracy
CR-5: Affordable price	EC-8: Manufacturing Cost	PC-1 through PC-8
CR-6: Easy to breathe through	EC-1: Mouthpiece Thickness EC-3: Resistance Valve Range EC-7: Device Weight	PC-1: Mouthpiece diameter PC-2: Wall thickness PC-4: Orifice area PC-7: Device mass
CR-7: Hygienic / easy to clean	EC-2: Material Softness EC-8: Manufacturing Cost	PC-3: Shore hardness PC-6: Housing wall thickness
CR-8: Durable for long use	EC-1: Mouthpiece Thickness EC-7: Device Weight EC-8: Manufacturing Cost	PC-2: Mouthpiece wall thickness PC-6: Housing wall thickness PC-7: Device mass

2.7.2. Design Gaps: Requirements Lacking Module Coverage

Gap 1: Easy Resistance Adjustment (CR-3)

Although EC-4 (Resistance Flexibility) strongly addresses adjustability, there is currently no dedicated part characteristic that directly measures the usability or ergonomics of the adjustment mechanism itself.

Current coverage: PC-4: Orifice cross-sectional area range; PC-5: Housing outer dimensions

Missing metrics:

- Adjustment dial torque or force
- Number of discrete adjustment positions
- Adjustment repeatability/tolerance

This creates a gap between the customer desire for intuitive real-time adjustment and the measurable physical characteristics currently defined in HOQ Level 2.

Gap 2: Hygienic / Easy to Clean (CR-7)

Hygiene and cleanability are only weakly linked to current part characteristics.

Current coverage: PC-3: Mouthpiece Shore hardness; PC-6: Housing wall thickness

Missing metrics:

- Surface roughness (Ra)
- Water resistance/sealing rating
- Removability/modularity for cleaning
- Antimicrobial material properties

Because the device is intended for repeated mouth contact during exercise, hygiene represents an important future refinement area.

Gap 3: Feedback Quality and User Experience (CR-4)

The current RTM captures pressure sensor accuracy but does not fully define the quality of the feedback experience itself.

Current coverage: PC-8: Pressure sensor accuracy

Missing metrics:

- App refresh/update rate
- Bluetooth latency
- Display readability
- Data storage capability

The current system captures measurement accuracy well, but software performance is not yet quantified through a dedicated PC.

2.7.3. Scope Creep: Part Characteristics with Weak Requirement Linkage

PC-6: Housing Wall Thickness

Housing wall thickness contributes to durability and manufacturing cost. However, it has limited direct influence on the highest-priority customer requirements identified in the conjoint analysis, such as comfort, feedback, and affordability.

Conclusion: Important structurally; weak direct customer linkage; should remain optimized primarily through structural analysis rather than customer preference weighting.

Potential Electronics Expansion

The current architecture anticipates future app-connected functionality, BLE communication, and rechargeable electronics. While these features improve the 'full package' perception described by interview participants, excessive expansion of app functionality could introduce unnecessary scope beyond the core breathing-training problem.

Examples of potential future scope creep (intentionally excluded from current prototype scope):

- Advanced fitness analytics
- Social connectivity (like Strava)
- Cloud-based data storage

2.7.4. Item Numbering System

Unique IDs for all functional modules are cross-referenced across HOQ L1, HOQ L2, product architecture, and BOM. See Section 2.5.2 for the clustered module layout with item numbers.

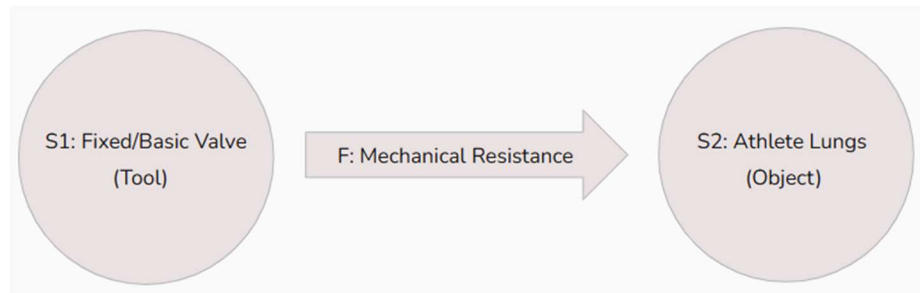
2.8. Use of TRIZ Throughout Iterations

2.8.1.1. Functional Decomposition

Functional Module	Primary Function	Supporting Engineering Characteristics
FM-1: Mouthpiece Module	Creates airtight and comfortable user interface	EC-1, EC-2
FM-2: Resistance & Adjustment Module	Generates and controls breathing resistance	EC-3, EC-4
FM-3: Sensor Module	Measures breathing pressure and airflow effort	EC-5
FM-4: Electronics & Feedback Module	Processes and transmits user feedback	EC-6
FM-5: Structural Housing Module	Protects system and minimizes size/weight	EC-7, EC-8

2.8.1.2. Substance Field Analysis

Initial System:



Substances: S1: Fixed resistance breathing device; S2: Athlete respiratory system

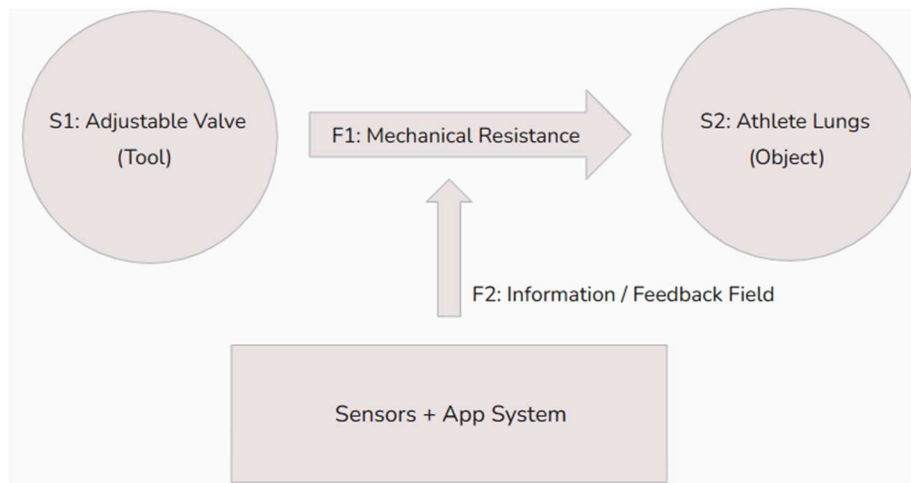
Field: F1: Mechanical airflow resistance

Problems Identified:

- Resistance is fixed or poorly adjustable
- Devices are uncomfortable during movement
- Resistance varies inconsistently with jaw position and leakage
- No meaningful feedback on training performance
- Devices interrupt normal training routines

This system was classified as an ineffective complete system, because although the resistance mechanism exists, it does not fully satisfy user needs.

Improved System:



Substances: S1: Adjustable threshold resistance mechanism; S2: Athlete respiratory system; S3: Sensor and feedback electronics

Fields: F1: Mechanical resistance field; F2: Information/feedback field

Improvements Added:

- Adjustable resistance wall for variable training intensity
- Threshold flap to maintain more consistent resistance behavior
- App-connected sensor system for measurable feedback
- Compact mouth-held geometry for natural integration into workouts

This transformed the design into an effective complete system by introducing both controllable resistance and information feedback.

2.8.1.2.1. Engineering Contradictions Being Solved

Contradiction 1: High Resistance vs Comfortable Breathing

Problem: Increasing resistance improves respiratory muscle training effectiveness but makes breathing less natural and less comfortable.

TRIZ Conflict: Improve: Training effectiveness | Worsens: User comfort and usability

Solution:

- Adjustable resistance wall
- Threshold flap providing consistent baseline resistance
- Multiple resistance levels for progressive training

Contradiction 2: More Features vs Affordable Price

Problem: Adding sensors, Bluetooth, and app feedback improves product value but increases manufacturing cost.

TRIZ Conflict: Improves: Functionality and feedback | Worsens: Cost and manufacturability

Solution:

- Compact low-power BLE electronics
- Simple pressure sensing instead of complex airflow sensing
- Injection molded modular housing
- Use of commercially available sensor and battery modules

Contradiction 3: Durability vs Lightweight Design

Problem: A stronger housing improves durability during athletics but increases device mass and discomfort.

TRIZ Conflict: Improve: Structural strength | Worsens: Wearability

Solution:

- Flexible silicone outer geometry
- Protective chamber design
- Compact internal electronics layout
- Localized reinforcement instead of globally thick walls

Contradiction 4: Tight Sealing vs Ease of Cleaning

Problem: A tightly sealed mouthpiece improves airflow consistency but makes cleaning more difficult.

TRIZ Conflict: Improve: Airtight performance | Worsens: Hygiene and maintenance

Solution:

- Modular architecture
- Smooth internal flow geometry
- Silicone mouthpiece with minimal crevices

2.8.1.2.2. Historical Solutions

Historical respiratory training solutions include:

- Handheld inspiratory muscle trainers
- Restrictive training masks
- Fixed-orifice breathing devices

These products generally suffer from poor comfort during exercise, bulkiness, lack of feedback, and limited adjustability. Most solutions require athletes to perform respiratory training separately from their normal workouts, reducing long-term user adoption.

2.8.1.2.3. Proposed Solution

The proposed solution combines:

- Adjustable mechanical resistance
- Threshold resistance flap
- Compact mouth-held form factor
- Integrated app-connected feedback

The product is specifically designed to allow respiratory training during normal athletic activity without significantly disrupting movement, comfort, or usability.

Key innovations include:

- Dual resistance mechanism (threshold flap + adjustable wall)
- Compact wearable geometry
- Integrated feedback system
- Modular construction

2.10. Traceability Matrix

CR	EC	PC	FM	BOM
CR-1 Comfortable mouth fit	EC-1, EC-2, EC-7	PC-1, PC-2, PC- 3, PC-7	FM-1	BOM-1
CR-2 Adjustable resistance	EC-3, EC-4	PC-4	FM-2	BOM-2, BOM- 3
CR-3 Easy adjustment	EC-4	PC-4, PC-5	FM-2	BOM-3
CR-4 Improvement feedback	EC-5, EC-6	PC-8	FM-3, FM-4	BOM-4, BOM- 5
CR-5 Affordable price	EC-8	PC-1 to PC-8	FM-1 to FM- 5	All BOM items
CR-6 Easy breathing	EC-1, EC-3, EC-7	PC-1, PC-2, PC- 4, PC-7	FM-1, FM-2	BOM-1, BOM- 2
CR-7 Hygiene	EC-2, EC-8	PC-3, PC-6	FM-1, FM-5	BOM-1
CR-8 Durability	EC-1, EC-7, EC-8	PC-2, PC-6, PC- 7	FM-5	BOM-1

2.11. Mechanical Analysis

2.11.1. Analytical

Due to the silicone structure of our product, most of our product will not need to be analyzed for fracture. However, analysis will need to be done to ensure the safety of the pressure sensor, BLE module, and battery. Our product will not be exposed to any force that could cause permanent deformation to silicone.

2.11.2. FEA

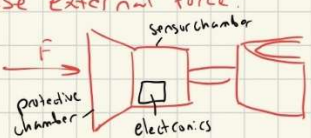
Finite element analysis is not necessary as no force will be applied to the product that is nearly strong enough to permanently deform or fracture the silicone. Only static analysis needs to be done to ensure that the pressure sensor is not broken. Ansys could be used to understand the deformation better of the sensor chamber, but a high factor of safety will ensure that the way it deforms will not affect the sensor, BLE module or battery.

2.11.2.1. Static

Because of the use of our product, any meaningful impact will be due to collision from the athlete into another athlete or an object. This load would travel through the product from the protective chamber into the walls of the sensor chamber and to the airway and the mouthguard. The threshold flap helps protect our electronics as almost no force can go through it as it is made to deform. When loading at the protective chamber, the only cause of damage to the electronics is when all 4 walls of the sensor chamber would buckle simultaneously, squeezing the electronics. Otherwise, load passes alongside the electronics instead of through them.

Any load on the top, bottom or side of the sensor chamber would be very small as the only realistic load would be from dropping the product. The silicone will effectively handle this load, especially with the mouthguard or protective chamber yielding the bulk of it as they are taller in every direction.

Worst case external force:

Load Path: 

Assuming a collision impulse with an athlete moving at conservative 10 m/s and amalgamating the buckling of all 4 sensor chamber walls at once.

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

$L = 12 \text{ cm}$

$$I = \frac{b^3 h^3}{12} - \frac{b_1^3 h_1^3}{12} = 0.33 \text{ cm}^4$$

$$I = \frac{(2.75+2t)(1+2t)^3 - (2.7)(1)^3}{12}$$

$P_{cr} = \frac{\pi^2 (2 \text{ MPa}) I}{(0.5)(12 \text{ cm})^2}$

Using a factor of safety of 1, protective chamber will absorb some load so even higher and already using conservative V_f :

Using a conservative impulse time of 0.01 s:

And a head-guard effective impact mass of 5 kg:

$$F_{max} = ma = (5 \text{ kg})(1000 \frac{m}{s^2}) = 5 \text{ kN}$$

Using $V_f = V_i + at$

$0 = 10 \frac{m}{s} + a(0.01)$

$a = -1000 \frac{m}{s^2}$

$5000 \text{ N} = (5.493 \times 10^{10} \frac{N}{m^4})(I)$

$9.12 \times 10^{-7} \text{ m}^4 = I$

$I = 0.912 \text{ cm}^4$

$0.912 \text{ cm}^4 = \frac{(2.75+2t)(1+2t)^3 - (2.7)(1)^3}{12}$

$t = 2.99 \text{ mm}$

Using $t = 3 \text{ mm}$, our product will have FS comfortably over 1

Threshold Flap Thickness calculation

to activate at 20% average athlete MIP:

$P = 0.2 \cdot 125 \text{ cmH}_2\text{O}$

$P = 25 \text{ cmH}_2\text{O} = 2.45 \text{ kPa}$

Cantilever Beam deflection:

$\Delta h = \frac{ML}{EI}$

$M = \frac{PbL^2}{6}$

$\Delta h = \frac{PL^3}{6EI} = \frac{PL^3}{6E \frac{bt^3}{12}} = \frac{2PL^3}{Et^3}$

$\frac{2(2450 \text{ Pa})(0.007 \text{ m})^3}{(2 \text{ MPa})(t^3)} = \frac{\pi}{2}$

$t = 0.812 \text{ mm}$

$I = \frac{bt^3}{12}$

$L = 7 \text{ mm}$

$b = 21 \text{ mm}$

2.11.2.2 Cyclic Loading

Our factor of safety for cyclic loading for our threshold flap is well above 1 showing our product is made to withstand many years of constant use.

Flap Cyclic Loading:

N = inhale-exhale closing motion
assuming - 40 breath per minute
- 60 min use per day
- 300 days per year
- 2 years usage

$N = 40 \cdot 60 \cdot 300 \cdot 2 = 1.44 \times 10^6 \text{ cycles}$

$$\epsilon_{max} = \frac{\delta}{2R} = \frac{\frac{F\pi}{4L}}{2} = \frac{.812 \text{ mm} \pi}{4(7 \text{ mm})}$$
$$R = \frac{L}{\theta} = \frac{L}{\epsilon}$$
$$\epsilon_{max} = 0.091$$
$$\sigma_{cyclic} = E \epsilon_{max} = (2 \text{ MPa})(0.091)$$
$$= 182.2 \text{ kPa}$$
$$\sigma_n = \sigma'_s (2N)^{-1/4}$$
$$\sigma_n = (2 \text{ MPa})(2.144 \times 10^6)^{-1/4}$$
$$\sigma_n = 451.95 \text{ kPa}$$
$$FS = \frac{\sigma_{allowable}}{\sigma_{cyclic}}$$
$$FS = 2.43$$

3. Market Identification

3.1. Target Market

Student and recreational endurance athletes in the United States.

3.1.1. Market Size – IBISworld/Other Market Reports

The U.S. athletic and sporting goods market is valued at approximately \$37.6 billion in 2025 and is predicted to reach \$60.1 billion in 2032. (Source 1 in Bibliography)

3.2. Consumer Behavior

3.2.1. Sociological Research

Research on wearable fitness technologies suggests that users are more likely to consistently engage with training devices that integrate naturally into existing exercise routines while providing trackable performance feedback. As fitness culture becomes increasingly data-driven, athletes and recreational users like tools that offer convenient data tracking without disrupting normal training flow. A great example of this is Fitbit, which is a great asset to runners and athletes due to its metrics and convenience. This trend supports the development of mouth-held, app-connected, respiratory training systems that combine performance monitoring with everyday usability. (Source 2)

3.2.2. Historical Solutions to Stated Problem

Historical approaches to convenient respiratory muscle training have included handheld breathing devices, restrictive breathing masks, and simple mouth-held airflow restricting devices. While these devices overall do help to improve endurance, they all fall short in one way or another such as bulkiness, limited adjustability, unnatural training experience, and lack of improvement feedback. Recent consumer demand has shifted toward undisruptive, adjustable, and feedback training systems that can integrate naturally into workouts and help users track their progress.

3.3. House of Quality (Again)

3.3.1. Focus on Competitor Analysis

3.3.1.1. HOQ Level 1

Customer Requirements	POWERbreathe Plus (1–5)	The Elevation Training Mask 2.0 (1–5)	Airofit Pro 2.0 (1–5)
Comfortable mouth fit	2	3	1
Adjustable resistance levels	1	4	5
Easy resistance adjustment	1	3	5
Clear improvement feedback	1	1	1
Affordable price	3	4	1
Easy to breathe through	2	5	3
Portable for athletes	1	4	2
Hygienic / easy to clean	4	3	3
Durable for long use	2	4	3

Scale: 1 = Poor, 3 = OK, 5 = Excellent

3.3.1.2. HOQ Level 2 (Develop BOM)

Part	Weight	Price
Injection Molded Parts (BOM-1 – BOM-3)	15.361 g	\$1.05
Pressure Sensor: Honeywell ABP2MDAN001PDSA3XX (BOM-4)	0.8–1.2 g	\$17.53
BLE Module: TI CC2340R52N0RKPR (BOM-5)	0.4–0.7 g	\$1.31
LiPo Battery Interface: MCP73831T-2ACI/OT (BOM-6)	0.005 g	\$0.76
Rechargeable Battery: TinyCircuits ASR00011 (BOM-7)	1.65–1.9 g	\$4.95
Total (per unit)	16.9 g	\$20.92

Injection molding cost for 10,000 units:

Cost Category	Amount
Material	\$554,902
Production	\$1,535,726
Tooling	\$8,376,507
Total	\$10,467,135
Cost Per Part	\$1.047

Note: BOM IDs for injection molded parts:

- BOM-1: Injection molded silicone housing
- BOM-2: Threshold Flap
- BOM-3: Adjustable resistance wall

3.4. Bass Model Forecast

3.4.1. Analogous Product or Historical Data for p (Innovation) and q (Imitation) Values

Based on the novelty and similarity of our design to other innovations in the fitness industry such as Fitbit, we estimated an innovation coefficient of $p = 0.02$ and an imitation coefficient of $q = 0.35$. Our product is not drastically revolutionary and targets a niche performance-oriented fitness market, so we chose a lower innovation coefficient. However, since expansion spreads rapidly, often through social influence in the fitness industry, we believe that growth will be exponential once innovators begin to show results with the product and encourage imitation, hence the higher imitation coefficient.

3.4.2. Projected Adoption Curve and Market Penetration

The projected adoption pattern follows a standard S-curve, with initially low market penetration during product introduction, followed by accelerated growth as awareness, user reviews, and social influence increase within fitness communities. Adoption is expected to expand through student athletes, recreational athletes, and social media before gradually approaching market saturation over time.

4. Financial Requirements

4.1. Financial Executive Summary

4.1.1. Price

\$100 per product

4.1.2. Expected Sales

Assuming 1,000 units in the first year, then doubling annually over the five-year period = 31,000 estimated sales = \$3,100,000 in revenue over 5 years.

4.1.3. Cost

\$20.92 per unit estimate

4.1.4. Net Present Value over 5 Years of Sales

Fixed cost to produce (tooling, etc.): \$50,000 estimate which includes initial tooling/molds, prototype iteration/testing, app/software setup and maintenance.

COGS (from BOM): \$20.92 per unit estimate.

Annual net: (price – cost) × quantity

Using average sales per year over 5 years for quantity: $(\$100 - \$20.92) \times (31,000 / 5) = \text{Annual net} = \$490,296$

Estimated Weighted Average Cost of Capital (5% default): 5%

Compounded annually:

Using 5% WACC, Sum of $(\text{Annual net} / (1 + \text{WACC})^n)$ over $n = 5$ years, minus Fixed costs = NPV

Net Present Value over 5 years = \$1,953,873

4.2. Pricing Policy

4.2.1. Derived from Conjoint and Other Observations

The target price of \$100 was derived from the second conjoint test results, which showed that customers were willing to pay up to \$100 for a product with mechanically adjusted resistance, app-connected feedback, and a variety of sizes.

4.3. Production Costs

4.3.1. Manufacturing Process Selection

4.3.1.1. Economic Batch Size

For low-volume prototyping (<100 units), the product is manufactured using TPU/SLA 3D printing and manual post-processing and assembly. This minimizes tooling costs, design iteration time, and prototype lead time. However, for medium-to-large production runs (>5,000–10,000 units), injection molding becomes significantly more economical because per-part cycle times are low, material waste is minimized, tooling cost is amortized over large quantities, and surface finish consistency improves substantially. The projected initial production target of approximately 10,000 units supports transitioning to injection molding.

4.3.1.2. Surface Roughness

The product requires smooth mouth-contact surfaces, comfortable airflow channels, minimal crevices for hygiene, and consistent sealing surfaces. Injection molding provides superior surface finish repeatability, dimensional consistency, and smooth airflow surfaces compared to additive manufacturing.

Target surface finish: SPI-B2 equivalent or smoother on mouth-contact surfaces; smooth internal airflow surfaces to reduce turbulence and improve cleanability.

3D printing was considered insufficient for final production because visible layer lines reduce comfort, increase bacterial retention, and increase post-processing requirements.

4.3.1.3. Part Architecture

The device geometry is well suited for injection molding because most features follow a consistent mold pull direction, the housing geometry is relatively shallow, the mouthpiece contains smooth radii and draft-friendly geometry, and internal airflow passages can be produced using simple core features.

The design intentionally minimizes undercuts, side actions, and complex sliding cores to reduce tooling complexity and mold cost.

The modular architecture also improves manufacturability: mouthpiece section can use softer silicone/TPU materials; housing section can use stiffer ABS or polypropylene; separate molding reduces multi-material tooling complexity.

The final assembly is completed using adhesive bonding (epoxy), mechanical snap features, or press-fit component integration, depending on manufacturing iteration.

4.3.2. Tooling Parameters and Costs for Selected Process

The selected high-volume manufacturing process is single-cavity injection molding using aluminum prototype tooling transitioning to hardened steel tooling for large-scale production.

4.3.2.1. CustomPartNet.com Tooling Parameters

Estimated Tooling Parameters:

Parameter	Estimated Value
Mold Type	Single-cavity aluminum mold
Material	TPU / Silicone-like thermoplastic elastomer
Part Volume	~15–20 g
Projected Production Volume	10,000 units
Draft Angle	1–2° minimum
Wall Thickness	2–4 mm nominal
Cycle Time	~30–60 seconds
Injection Pressure	Moderate
Machine Clamp Force	~20–50 ton class

Estimated Tooling Cost:

Tooling Component	Estimated Cost
Aluminum prototype mold	\$3,000 – \$8,000
Production steel mold	\$15,000 – \$30,000
Mold design & DFM review	\$2,000 – \$5,000
Initial sampling/testing	\$1,000 – \$3,000
Total estimated initial tooling investment	~\$20,000 – \$40,000

Per-Part Manufacturing Estimate:

Cost Category	Estimated Cost per Unit
Injection molded polymer parts	~\$1.05
Pressure sensor	~\$17.53
BLE module	~\$1.31
Battery interface	~\$0.76
Rechargeable battery	~\$4.95
Assembly + packaging	~\$3–5
Total estimated COGS	~\$28–31 per unit

4.3.2.2. Quotes from Manufacturer / Manufacturing Considerations

Because the product architecture is relatively simple and uses mostly thin-walled polymer geometries, the manufacturing process is considered favorable for outsourced injection molding. The design intentionally reduces side actions, complex undercuts, and multi-shot molding requirements, which lowers tooling cost and simplifies manufacturing.

The manufacturing-heavy aspects of the design include: soft-touch mouthpiece production, thin threshold flap geometry, tight tolerance airflow paths, and electronics integration and sealing.

The team anticipates using outsourced injection molding, commercial off-the-shelf electronics, and manual final assembly during early production stages to minimize capital investment and simplify scaling.

Several manufacturers and industry sources indicate that simple single-cavity molds for small polymer products commonly fall within the \$3,000–\$15,000 range for aluminum tooling, while hardened steel production molds can range from \$20,000–\$100,000 depending on complexity and production life.

4.4. Capital Investment Required

4.4.1. How Will You Raise Money for Manufacturing Equipment or Afford to Outsource?

Our company will raise the money to create our product mainly through early-stage funding such as university entrepreneurship grants or small-scale angel investment. From there, we can outsource injection molding, electronics fabrication, and assembly to reduce the upfront capital required for tooling and production equipment, which would allow us to start making revenue and establish a strong brand name.

4.4.2. How Will You Raise Money to Pay Employees to Scale Up Production?

In order to scale up and be able to pay employees, we would likely seek venture capital investment and trade partial equity for funding. Once we secure further funding, we can start to manufacture our product in-house and build a team of experts necessary to help us expand.

5. Legal Requirements

5.1. Safety & Environmental Regulations for Your Product's Manufacturing

Polymer Processing: Injection molding and thermoplastic processing require proper ventilation, temperature control, and worker PPE to reduce exposure to VOCs, heated polymer fumes, and particulate emissions. Manufacturing facilities are expected to comply with OSHA workplace safety standards, EPA air quality regulations, and local industrial ventilation requirements.

Electronics Manufacturing: Electronics integration introduces soldering, battery handling, and PCB assembly processes. Manufacturing must comply with the European Union's RoHS Directive, which restricts hazardous substances such as lead, mercury, cadmium, and certain flame retardants.

Battery Safety: The rechargeable lithium battery system must comply with UN38.3 transportation and handling standards, along with overcharge and short-circuit protection requirements. These standards help ensure the battery remains safe under vibration, thermal cycling, impact, and other conditions relevant to athletic use.

Environmental Regulations: The manufacturing process is designed to minimize polymer waste, packaging waste, VOC emissions, and energy consumption. Injection molding was selected partly because it produces relatively low material waste and supports recyclable thermoplastics. The design also aligns with RoHS and WEEE principles intended to reduce hazardous electronic waste and improve recyclability throughout the product lifecycle.

5.2. Safety & Environmental Regulations for Your Product's Use

Airflow Restriction Safety: The resistance system is designed to maintain breathable airflow at all resistance levels and prevent complete airway blockage. The threshold flap and adjustable resistance wall provide training resistance without fully sealing airflow, while still allowing emergency airflow release if excessive pressure is applied.

Material Safety: Mouthpiece materials must be non-toxic, skin-safe, mouth-safe, and resistant to degradation from repeated saliva exposure. Preferred materials include medical-grade silicone and food-safe thermoplastic elastomers (TPE/TPU). Materials restricted under RoHS are avoided due to their environmental and health risks.

Electrical Safety: The embedded electronics operate at low voltage and are isolated within the protective housing to minimize moisture intrusion and short-circuit risk. The charging system incorporates overcharge protection, battery management circuitry, and short-circuit protection.

Athletic Use Safety: The compact mouth-held geometry was designed to reduce injury risk during movement and minimize protrusions during contact sports. The soft silicone geometry also reduces oral injury risk and hard-edge impacts with athletes or equipment.

5.3. How Will the Products Reduce Environmental Impact on Production?

The product geometry was optimized to minimize wall thickness where structurally acceptable, reduce device mass, and lower polymer consumption per unit. The compact architecture also reduces raw material usage, shipping weight, and packaging volume. Injection molding was selected because it produces low material waste, supports recyclable thermoplastics, and achieves fast, energy-efficient cycle times at scale. Compared to subtractive manufacturing methods, injection molding generates significantly less scrap material. The use of commercial off-the-shelf electronics also reduces custom PCB waste and improves manufacturing consistency.

5.4. How Will the Products Reduce Environmental Impact Upon End of Life?

Where possible, the housing will use recyclable thermoplastics and silicone or TPU materials with accessible recycling pathways. Although mixed-material systems present recycling challenges, the modular design improves recyclability compared to fully integral products. The rechargeable battery must be disposed of according to local lithium battery recycling regulations and electronic waste handling standards.

5.5. Potential Liability Issues for Use

5.5.1. Prediction of Misuse Cases

Excessive Resistance Setting: A user may adjust the device to unsafe resistance levels, potentially causing dizziness, hyperventilation, or breathing distress. This risk is mitigated through mechanical resistance limits, warning labels, and graduated training recommendations.

Use During Unsafe Activities: A user may wear the device during swimming, high-contact competition, or while operating a vehicle, potentially impairing breathing during emergencies. This is addressed through activity restriction guidance and cautionary labeling.

Failure to Clean the Device: Repeated use without cleaning could lead to bacterial growth, odor, and material degradation. The smooth silicone geometry and modular construction support easier cleaning, while product documentation includes cleaning instructions and recommended maintenance intervals.

Continued Use After Structural Damage: A user may continue using the device after damage to the threshold flap or housing, risking inconsistent resistance, detached components, or electronics exposure. Mitigation measures include inspection recommendations and durable silicone geometry designed to fail gradually rather than catastrophically.

Improper Charging: Using incompatible chargers or exposing the charging port to moisture could result in overheating or electrical failure. The design includes charging protection circuitry, protected charging geometry, and safety warnings in the user documentation.

Use by Individuals with Respiratory Conditions: A user with asthma, COPD, or another respiratory condition could experience complications if using the device without medical guidance. The product therefore includes non-medical-use disclaimers and warnings advising users with respiratory conditions to consult a physician before use.

5.6. Failure Mode and Effect Analysis

5.6.1. Identify Potential Failure Modes, Effects, and Severity

Failure Mode	Effect	Severity	Occurrence	Detection
Silicone permanent deformation	Trainer will not work as expected; resistance levels will not provide expected resistance	6	2	7
Electronics damage	Pressure sensor will no longer work and the trainer will not give electronic feedback	10	7	1
Flap Cyclic loading failure	The resistance will rip off a bit or a lot and lose its intended function as it will bend too easily	9	1	2
Flap thickness incorrect	The flap will not bend at the correct pressure	7	3	5

5.6.2. Prediction of Misuse Cases

The only plausible misuse of our product would be if the user doesn't wrap their mouth around the airway and their teeth correctly in the mouthguard, resulting in them breathing around the device and no longer receiving the resistance they should be.

5.6.3. Risk Priority Numbers (RPN) and Mitigation Strategies

Failure Mode	Risk Priority Number (RPN)	Mitigation Strategies
Silicone permanent deformation	84	Try different types of silicone to find the best one and ensure no permanent deformation.
Electronics damage	70	Increase the thickness of the sensor chamber walls to reduce the chance of buckling.
Flap Cyclic loading failure	18	Not much to mitigate as it is extremely unlikely and only happens after years and years of use.
Flap thickness incorrect	105	Make a lot of prototypes to ensure the right thickness. Use precise manufacturing technique to ensure no faults.

5.6.4. Risk Mitigation to Lower RPN

See RPN mitigation strategies in Section 5.6.3 above.

5.7. Industry Standards and Certification

5.7.1. Relevant Industry Standards (FMVSS, UL, ISO, etc.)

ISO 10993 – Biocompatibility of Medical Materials: Because the mouthpiece directly contacts the user's teeth, gums, saliva, and soft tissue during use, material selection should follow ISO 10993 guidance for biocompatibility, cytotoxicity, irritation potential, and overall material safety during prolonged human contact. Medical-grade silicone and food-safe TPU materials are preferred to minimize health risks associated with repeated oral exposure.

RoHS – Restriction of Hazardous Substances: The product incorporates sensors, BLE electronics, a rechargeable battery, and charging circuitry, requiring compliance with RoHS restrictions on hazardous substances such as lead, mercury, cadmium, and restricted flame retardants. RoHS compliance improves environmental sustainability, export compatibility, and consumer safety.

WEEE – Waste Electrical and Electronic Equipment Directive: Because the product contains embedded electronics and a rechargeable battery, end-of-life disposal and recycling are important regulatory considerations. WEEE principles support battery removal, electronics recycling, and reduction of electronic waste. The modular architecture of the product supports partial disassembly and component-level recycling.

UL Standards for Rechargeable Battery Systems: The device uses a rechargeable lithium battery and integrated charging circuitry. This includes UL2054 (Household and Commercial Batteries) which addresses battery overheating, charging safety, short-circuit prevention, and electrical enclosure safety.

FCC Part 15 – Wireless Communication Compliance: The BLE module transmits wireless data to a companion smartphone application, meaning the product intentionally emits radio-frequency signals. FCC Part 15 compliance is therefore required for wireless consumer electronics sold in the United States to ensure acceptable RF emissions and reduced interference with other devices.

ASTM Standards for Consumer Products: Several ASTM standards related to polymer durability, consumer product safety, and athletic protective equipment may also be relevant depending on final commercialization direction. Although no ASTM standard specifically governs respiratory trainers for athletes, ASTM testing methods could support durability validation, cyclic loading verification, and impact resistance testing relevant to athletic use.

5.7.2. Certifications Required for Your Product Category

Because the device is intended as a consumer athletic training product rather than a medical device, the certification burden is lower than for clinical respiratory systems. However, several certifications and compliance pathways would still be required prior to commercialization.

Likely required certifications include:

- FCC Part 15 compliance for BLE wireless communication
- RoHS compliance for hazardous substance restrictions
- UN38.3 battery transportation compliance for lithium battery shipping and handling
- UL battery safety compliance for charging and electrical safety
- WEEE compliance for electronic waste and recycling requirements

Additional certifications that are beneficial but not strictly required include:

- ISO 10993 material testing for consumer safety validation
- ASTM durability testing for reliability verification
- CE marking for sales in European markets

Medical Device Classification: The product is not intended to diagnose disease, treat respiratory conditions, or provide medical therapy. It is therefore intended to remain within the category of consumer fitness equipment or wearable athletic technology rather than FDA-regulated medical equipment. To maintain this classification, marketing materials must avoid therapeutic claims, medical treatment language, or disease prevention claims that could unintentionally trigger FDA medical device classification and substantially increase regulatory requirements.

5.8. Intellectual Property Considerations

5.8.1. Pursue Utility, Design Patents? Why?

A utility patent would protect the functional operation of the product. The strongest patentable concepts center on the resistance mechanism configuration, airflow path architecture, and integration of performance feedback with athletic respiratory training. Potential utility patent claims could include the adjustable threshold resistance system, the combination of the threshold flap and adjustable resistance wall operating in series, the compact athletic respiratory trainer geometry, and the integrated feedback-based training architecture.

Protecting these features through utility patents would help secure the core functionality of the product, strengthen competitive differentiation, and support future commercialization, licensing opportunities, and investor discussions.

A design patent would protect the visual and ornamental appearance of the product, including the mouthpiece geometry, external housing shape, and overall industrial design. The compact curved form factor and integrated mouthguard appearance create a recognizable product identity that is worth protecting.

Design patents are particularly valuable in the athletic wearables market because visual differentiation strongly supports branding, and competitors may otherwise imitate the external styling of a successful product without infringing on functional utility claims.

5.8.2. Trade Secrets? Why?

Certain aspects of the product may be more effectively protected as trade secrets rather than through patent filings, particularly where public disclosure would weaken competitive advantage without providing proportional legal benefit.

Potential trade secrets include:

- material formulation selections
- sensor signal processing algorithms
- app-side performance analysis logic

Trade secret protection is advantageous because it avoids public disclosure, has no expiration date, and can protect manufacturing know-how or algorithmic methods more effectively than patents in areas where reverse engineering is difficult.

However, trade secret protection becomes more difficult once the product enters the market and is subject to teardown or reverse engineering. For this reason, trade secrets are best reserved for internal calibration methods, software logic, and proprietary manufacturing processes that are not directly observable from the finished product.

5.8.3. Trademark? Describe



The Breathlete trademark includes both the product name and logo identity associated with the respiratory trainer. The name combines “breath” and “athlete” to reflect the product’s focus on athletic respiratory performance and endurance training. The logo uses a simplified aerodynamic airflow symbol and performance-oriented styling, and the slogan, “Train Every Breath. Elevate Every Performance.” further strengthens the product’s identity within the wearable fitness and athletic technology market. Trademark protection would help establish brand recognition and prevent competitors from using similar branding or visual identities

5.9. What Are Your Next Steps?

Next steps include creating a prototype, establishing the app-connected feedback system, and conducting additional athlete testing to evaluate comfort, resistance performance, and usability. We would also pursue patent and trademark filings, seek early-stage funding, and begin working with manufacturers to prepare for small-scale production and future commercialization.

6. Sources

6.1. Bibliography

- Source 1 (Market Size): <https://www.psmarketresearch.com/market-analysis/us-athletic-sporting-goods-market>
- Source 2 (Sociological): <https://pubmed.ncbi.nlm.nih.gov/31976079/>
- Source 3 (Tooling/Manufacturing Cost): <https://www.custompartnet.com/>
- Source 4 (Injection Molding Cost): <https://formlabs.com/blog/injection-molding-cost/>
- Source 5 (ISO 10993 Biocompatibility Standards): <https://www.iso.org/standard/68936.html>
- Source 6 (European Commission RoHS Directive): https://environment.ec.europa.eu/topics/waste-and-recycling/rohs-directive_en
- Source 7 (WEEE / RoHS Guidance): <https://www.trade.gov/eu-weeerohs>
- Source 8 (FCC Wireless Device Regulations): <https://www.fcc.gov/wireless/bureau-divisions/technologies-systems-and-innovation-division/rules-regulations-title-47>
- Source 9 (UL2054 Standards): https://www.lipolbattery.com/UL2054_Certification_of_Lithium-ion_Battery.html?srsId=AfmBOorPf5ASQxAXpCo5ql_2AVJjIxHv9uDRNctUThJNTyYriWhyDmqq
- Source 10 (ASTM International Standards): <https://www.astm.org>